

Ayush Mondal

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Education

RCC Institute of Information Technology (RCCIIT) <i>B.Tech in Electronics and Communication Engineering (7.3)</i>	<i>2022 – 2026</i> <i>Kolkata, WB</i>
Purwanchal Vidya Mandir <i>Secondary and Higher Secondary Education (90/75)</i>	<i>2022</i> <i>Kolkata, WB</i>

Projects

- AWS Data Pipeline (AWS S3, Lambda, Glue, EventBridge, SNS, CloudWatch, Python)** [GitHub](#)
- Developed a Python-based serverless ETL pipeline using AWS Glue and AWS Lambda to ingest, clean, and transform raw CSV data into analytics-ready outputs, implemented modular workflow components for maintainability and faster iteration, and improved data quality while reducing manual preprocessing effort.
- ResearchMind - Multi Agent Research Assistant (Python, LangChain, Google Gemini, BeautifulSoup Streamlit, Tavily)** [GitHub](#), [Deployed Link](#), [Demo Video](#)
- Built ResearchMind, a multi-agent research assistant developed with LangChain and Google Gemini to automate the research workflow from web search and content extraction to structured report generation and critique. Implemented a modular agent-based architecture with a Streamlit user interface and a command-line runner, enabling configurable, reproducible research pipelines driven by environment-based API keys and model settings.
- AI Video Assistant (Python, pydub/ffmpeg-python, Whisper (STT), sentence-transformers (embeddings), Chroma, MistralAI)** [GitHub](#), [Deployed Link](#), [Demo Video](#)
- Developed an end-to-end RAG-based Video Assistant using Python, FFmpeg, ChromaDB, and SQLite to enable semantic video search, speech-to-text transcription, and AI-powered summarization through a modular retrieval pipeline.

Work Experience

- Consistent Services**
Freelance (Ongoing) *Kolkata, WB*
- Developing an interactive 3D WAP-7 locomotive visualization system using Unity/Unreal Engine, simulating traction, braking, auxiliary and pneumatic systems with real-time animations, selectable components and exploded views, while integrating bilingual audio-visual training modules and adhering to RDSO/CLW technical standards.
- Allotrope Systems Private Limited**
Intern *Kolkata, WB*
- Developed a Hospital Management System backend using Node.js, Express.js, and MongoDB, implementing REST APIs and Mongoose schemas for core workflows (appointments, billing, beds, discharge), while applying MVC architecture, async/await, and centralized error handling for scalable and maintainable design.
- Smart India Hackathon**
Participant *Kolkata, WB*
- Developed a multilingual (English/Hindi/Bengali) voice-interactive chatbot using Web Speech API with integrated speech recognition and text-to-speech for real-time user interaction

Technical Skills

Languages: Python, SQL, JavaScript, Java
Frameworks/Libraries: Langchain, ,RAG Pipelines, AI Agents, Multi-Agent Systems, Express.js, LLM, FastAPI, PySpark, Mongoose, GenAI
Databases: MongoDB, VectorDB, chromaDB, PostgreSQL, MySQL
Tools: Git, GitHub, AWS, Postman, Ollama, Mistral, GeminiAPI

Certifications & Achievements

AMCAT (2025): Automata 93/100, Advanced Quant 84/100, Programming 79/100 — AWS Cloud Practitioner (GFG)
— Langchain : Foundations (Langchian Academy) — Intro to Modern AI (Cisco) — Intro to Python (Nasscom)